STARTUP PROFIT ANALYSIS · REGRESSION & POWER BI

Profit Analysis

R&D · Marketing · Administration · State · Regression · Power BI

PROJECT OBJECTIVE

Objective

Analyze how R&D, Marketing, Administration, and State impact startup profitability. Build a regression model, generate profit predictions, and surface actionable business insights through Power BI.

Six Core Goals

Analyze Spending Impact - Analyze the impact of R&D, Marketing, and Administration spending on startup profitability using regression analysis.

Build a Predictive Model - Develop a Multiple Linear Regression model to predict startup profit based on available financial data.

Evaluate Model Performance - Assess the model using key metrics — R^2, MSE, RMSE, and MAE — to ensure accuracy and reliability.

Generate Business Insights - Identify the most influential profit drivers and derive actionable insights for strategic investment decisions.

Support Data-Driven Decisions - Provide a decision-support framework to help management optimize resource allocation and improve financial result.

Visualize with Power BI - Create an interactive Power BI dashboard to visualize spending trends, profitability patterns, and regional performance.

DATASET OVERVIEW

Dataset Overview

50 Startups
Across 3 states

5 Features
Input + target variables

100% Validated
Clean & quality-checked

DATASET FEATURES

R&D Spend
Research & development investment

Administration
Admin & overhead costs

Marketing Spend
Sales & marketing expenditure

State
NY, California, Florida

Profit
Target variable (USD)

DESCRIPTION OF DATASET

Column	Type	Description	Role
RD Spend	Numerical	Amount invested in Research & Dev.	Independent Variable
Admin. Spend	Numerical	Administrative operating expenses	Independent Variable
Mar. Spend	Numerical	Marketing and advertising exp.	Independent Variable
State	Categorical	Location of startup (CA, FL, NY)	Categorical Predictor
Profit	Numerical	Net profit earned by the startup	**TARGET Variable**

ANALYTICAL METHODOLOGY

Regression Analysis Methodology

1. Data Extraction
Pull from MySQL
startup' table

2. Data Validation
Clean, deduplicate
& verify integrity

3. Regression Analysis
Multiple linear
regression model

4. Model Evaluation
R^2, RMSE &
residual analysis

5. Profit Prediction
Forecast 2 new
input scenarios

6.Power BI Dashboard
Interactive BI
visualizations

7. Insights & Recs
Strategic business
recommendations

Data Cleaning and Pre-processing

1. Missing Value Detection
Each column examined for NULL/missing values via SQL queries and Orange Data Table. No missing values found.

5. Range Validation
Min and max values of each numerical attribute reviewed to detect invalid or out-of-range entries.

2. Duplicate Record Validation
All startup records checked for duplicates. No duplicate observations were identified.

6. Outlier Assessment
Scatter plots and summary statistics examined. No extreme outliers requiring removal were observed.

3. Data Type Verification
RD_Spend, Administration, Marketing_Spend, and Profit confirmed as numerical; State confirmed as categorical.

7. Final Dataset Validation
Dataset confirmed to contain 50 complete observations — suitable for regression analysis.

4. Categorical Variable Handling
State cannot be used directly in linear regression. Orange internally converted State into numeric dummy/one-hot encoded indicators for California, Florida, and New York.

REGRESSION INFO

Regression Analysis

Modal Overview

Technique: Linear Regression

Target Variable: Profit (Dependent)

Predictor Variables: RD Spend, Administration, Marketing Spend, State

Tool Used: Orange Data Mining

R^2 Achieved: ≈ 0.944 (94.4%)

Evaluation Matrices

R^2 (R-Squared)
Measures proportion of variance in Profit explained by the model. Ranges from 0 to 1; higher is better.

RMSE (Root Mean Square Error)
Measures average prediction error in the same unit as Profit. Penalizes large errors more.

MSE (Mean Squared Error)
Average squared difference between actual and predicted profit. Used to check overall model fit.

MAE (Mean Absolute Error)
Average absolute prediction error. Easy to interpret in business terms.

Regression Assumptions

Linearity
A linear relationship exists between the predictor variables (RD Spend, Marketing, Administration, State) and Profit.

Independence of Observations
Each startup record represents an independent observation. The outcome of one startup does not influence another.

Normality of Residuals
Prediction errors are randomly distributed around the regression line — the residuals follow an approximately normal distribution.

No Multicollinearity
Predictor variables are not highly correlated with one another, ensuring each variable contributes independent information to the model.

Homoscedasticity
The variance of prediction errors remains approximately constant across different observations — ensuring consistent model reliability.

Model Limitation

Small Sample Size

The dataset contains only 50 startup observations, which limits the model's ability to generalize to broader startup populations or industries.

Limited Geographic Scope

The analysis includes startups from only three states — California, Florida, and New York. This restricts the model's applicability to startups in other regions or countries.

Excluded External Factors

External business factors such as market competition, customer demand, inflation, taxation, and macroeconomic conditions are not included in the model.

Important

The regression model should be used as a decision-support tool to estimate startup profitability. Final investment decisions should also consider external business and economic factors that are not included in the model.

Future Scope & Enhancement

Feature Selection

Apply statistical feature selection methods to identify only the most significant predictors. Removing irrelevant variables can improve model stability and reduce computational complexity.

Ridge Regression

Ridge regression adds an L2 regularization penalty to the model, which shrinks coefficients and reduces overfitting risk. Particularly useful when multicollinearity is present among predictor variables.

Lasso Regression

Lasso regression applies an L1 regularization penalty that can shrink some coefficients to exactly zero, effectively performing automatic feature selection. This can produce simpler, more interpretable models.

Important - These techniques may improve model stability, reduce overfitting risk, and test whether a simpler model performs better on larger datasets.

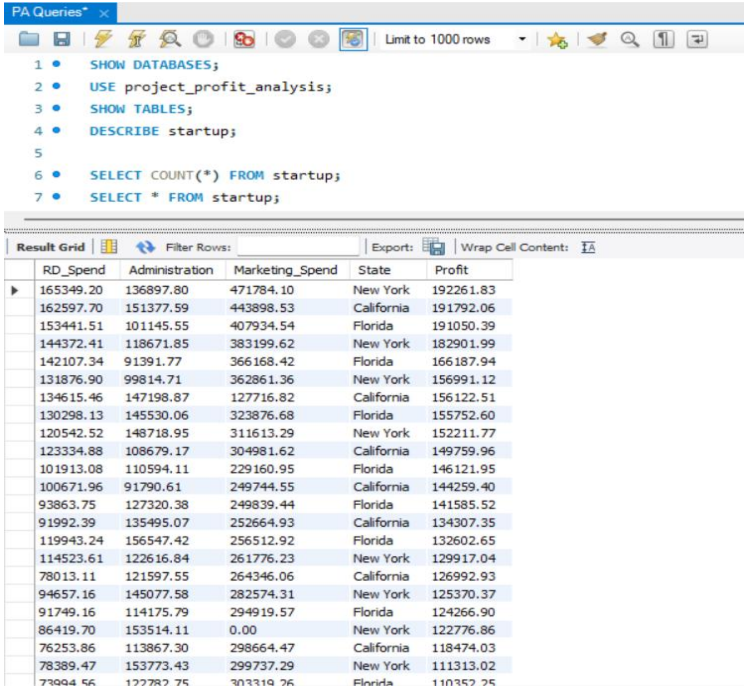
Data Extraction & Validation

Database Connection
Connected to the MySQL database 'project_profit_analysis' to access the startup table.

SQL Query Execution
Executed SQL SELECT queries to retrieve all 50 startup records with their five attributes.

Data Validation Check
Verified column names, data types, NULL values, and record count post-extraction.

Data Confirmation
Confirmed that the startup table contained 50 complete records: RD_Spend, Administration, Marketing_Spend, State, and Profit.




```
SHOW DATABASES;
USE project_profit_analysis;
SHOW TABLES;
DESCRIBE startup;

SELECT COUNT(*) FROM startup;
SELECT * FROM startup;
```

RD_Spend	Administration	Marketing_Spend	State	Profit
165349.20	136897.80	471784.10	New York	192261.83
162597.70	151377.59	443898.53	California	191792.06
153441.51	101145.55	407934.54	Florida	191050.39
144372.41	118671.85	383199.62	New York	182901.99
142107.34	91391.77	366168.42	Florida	166187.94
131876.90	99814.71	362861.36	New York	156991.12
134615.46	147198.87	127716.82	California	156122.51
130298.13	145530.06	323876.68	Florida	155752.60
120542.52	148718.95	311613.29	New York	152211.77
123334.88	108679.17	304981.62	California	149759.96
101913.08	110594.11	229160.95	Florida	146121.95
100671.96	91790.61	249744.55	California	144259.40
93863.75	127320.38	249839.44	Florida	141585.52
91992.39	135495.07	252664.93	California	134307.35
119943.24	156547.42	256512.92	Florida	132602.65
114523.61	122616.84	261776.23	New York	129917.04
78013.11	121597.55	264346.06	California	126992.93
94657.16	145077.58	282574.31	New York	125370.37
91749.16	114175.79	294919.57	Florida	124266.90
86419.70	153514.11	0.00	New York	122776.86
76253.86	113867.30	298664.47	California	118474.03
78389.47	153773.43	299737.29	New York	111313.02
73994.56	122782.75	303319.26	Florida	110352.25

Regression workflow in Orange

01. **File / SQL Input -** Imported the MySQL dataset into Orange using the SQL Table widget.

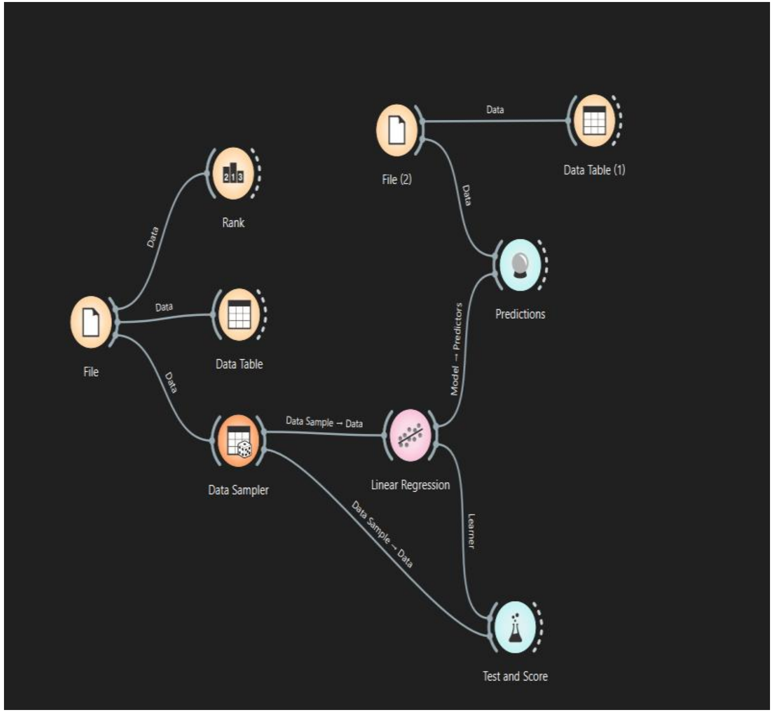
02. **Data Table -** Used the Data Table widget to inspect all 50 records and verify data completeness.

03. **Feature Ranking -** Applied the Rank widget to identify the most influential variables affecting Profit.

04. **Linear Regression -** Connected the Linear Regression widget, set Profit as target, remaining columns as predictors.

05. **Test & Score -** Applied 10-fold cross-validation in the Test & Score widget to evaluate model performance metrics.

06. **Predictions -** Used the Predictions widget to forecast Profit for new startup investment scenarios.



Model Performance Evaluation

R^2 Score

0.944

Excellent fit — model explains 94.4% of profit variance

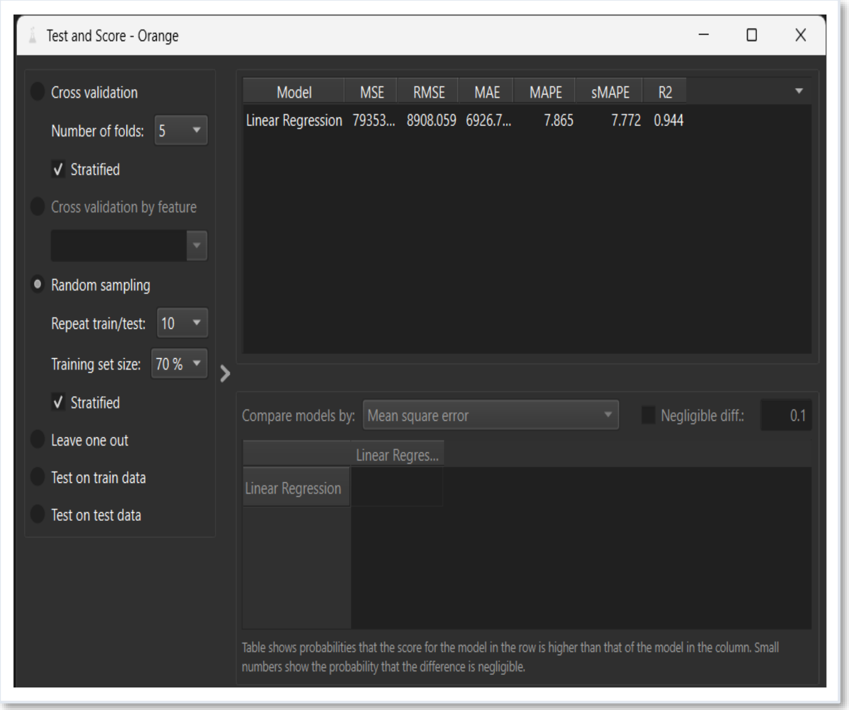
Prediction Accuracy

~94%

High reliability in estimating startup profit for new inputs

Model Interpretation

The regression model achieved an R^2 value of approximately 0.944, indicating that about 94.4% of the variation in profit can be explained by the selected variables — R&D Spend, Marketing Spend, Administration, and State. This demonstrates strong predictive performance and reliability, confirming that the model captures the key profit drivers effectively.



REGRESSION RESULTS

Profit Predication Results

0.944

R^2 Score

Model explains 94.4%of profit variance

$69,594

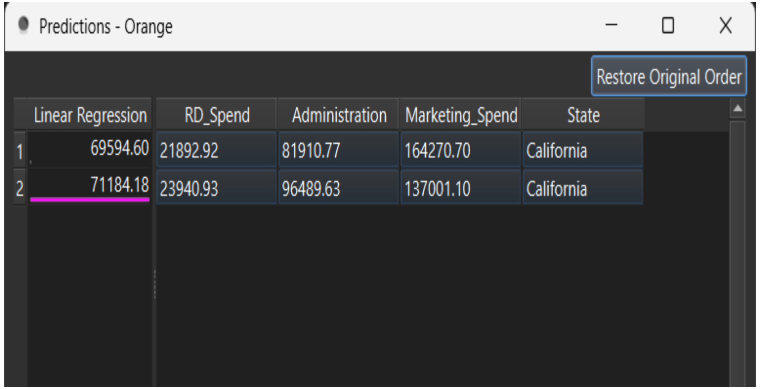
Predicted Profit 1

Lower R&D investment paired with high marketing results in a predicted profit of ~$69.6K.

$71,184

Predicted Profit 2

Slightly higher R&D investment with moderate marketing yields a better predicted profit of ~$71.2K.



	Linear Regression	RD_Spend	Administration	Marketing_Spend	State
1	69594.60	21892.92	81910.77	164270.70	California
2	71184.18	23940.93	96489.63	137001.10	California

INPUT SCENARIOS FOR PREDICTION

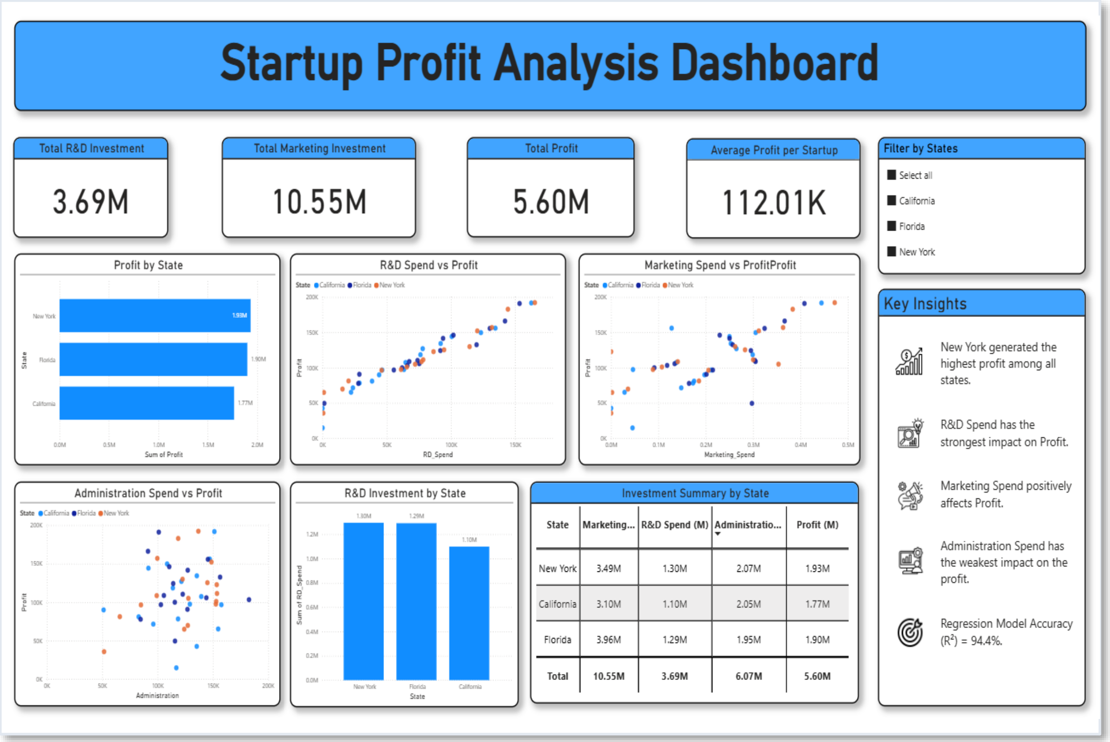
Startups	R&D Spend ($)	Administration ($)	Marketing ($)	Predicted Profit
Startup 1	21,892.92	81,910.77	164,270.70	$69,594.60
Startup 2	23,940.93	96,489.63	137,001.10	$71,184.18

Power BI Dashboard Overview

Dashboard Components

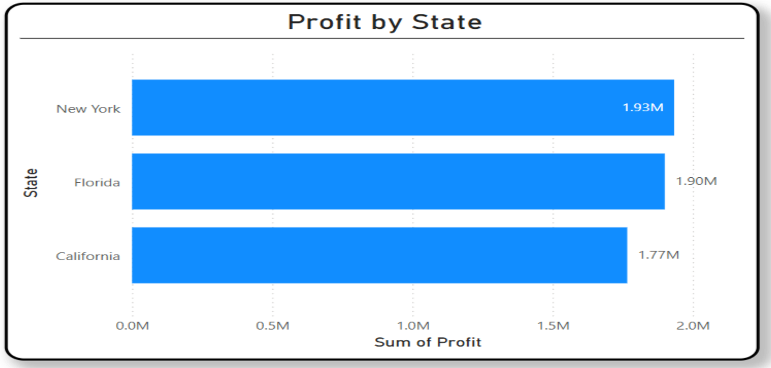
- **Profit by State -** Bar chart comparing state wise total profit
- **R&D vs Profit -** Scatter plot: strong positive correlation
- **Marketing vs Profit -** Scatter plot: marketing impact on profit
- **Administration vs Profit -** Scatter plot: weak relationship
- **State-wise R&D Investment –** Comparative bar chart across CA, FL, NY

Purpose : Enable quick exploration of key business metrics, spending patterns, and profitability trends for data-driven decisions.



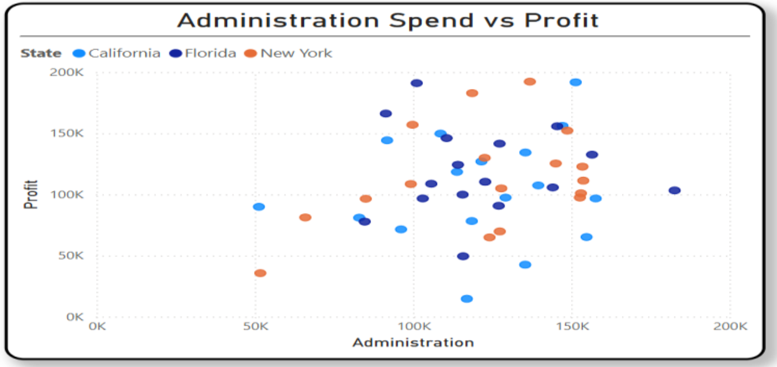
Dashboard Charts and Visualizations – I

Profit by State



Purpose: Compare total profit across CA, FL, NY
Obs: New York: highest profit - California: lowest in dataset

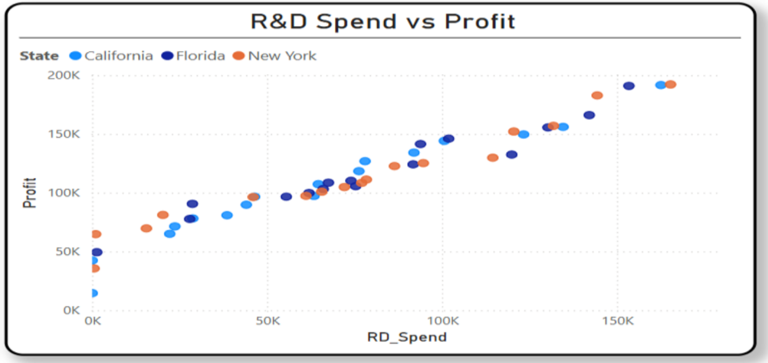
Administration Spend vs Profit



Purpose: Evaluate admin spending impact on profit
Obs: Weak/no consistent relationship - scattered data

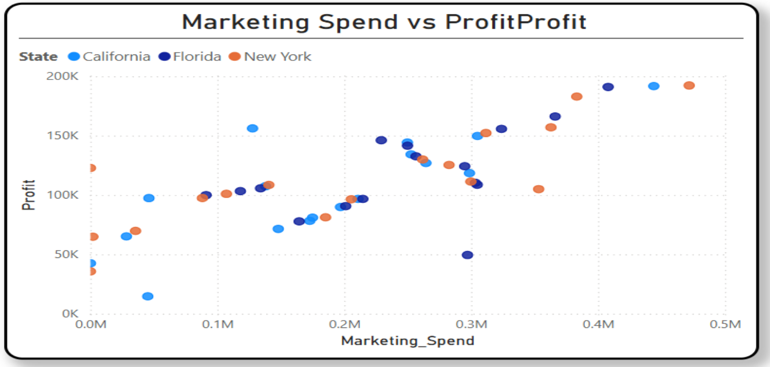
Dashboard Charts and Visualizations – II

R&D Spend vs Profit



Purpose: Show relationship between R&D investment and profit
Obs: Strong positive correlation — more R&D = higher profit

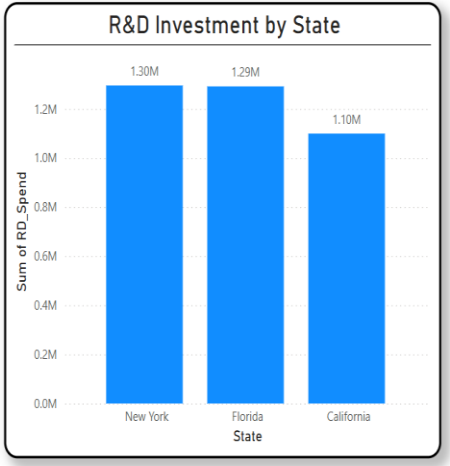
Marketing Spend vs Profit



Purpose: Analyze marketing expenditure vs profitability
Obs: Moderate positive relationship observed

Dashboard Charts and Visualizations – III

State-wise R&D Investment



Purpose: Compare R&D spending across states
Obs: NY and FL record highest R&D; CA comparatively lower

Business Meaning — Key Takeaways

- NY highest total profit; benchmark its startup practices.
- R&D budget is the most impactful investment lever.
- Marketing ROI must be tracked before further scaling.
- Admin costs offer limited return — optimize spend.

Key Business Insights

Top Performing State - New York generated the highest cumulative profit among the three states, indicating stronger overall startup financial performance in the dataset.

Model Performance - The regression model achieved an R^2 value of approximately 0.944, meaning about 94.4% of the variation in startup profit is explained by the selected predictor variables.

Marketing Positive Impact - Marketing Spend showed a moderate positive relationship with Profit. Although its impact was lower than R&D, increased marketing expenditure generally contributed to improved profitability.

Weakest Impact - Administration Spend displayed the weakest relationship with Profit. Higher administrative expenses did not consistently result in increased profit, suggesting limited financial returns from additional admin spending.

Strongest Driver - R&D Spend exhibited the strongest positive relationship with Profit. Startups investing more in R&D consistently achieved higher profit, making it the most influential investment category.

Predictive Accuracy - The prediction results demonstrated practical usefulness — startup profit can be estimated for new investment scenarios before actual business decisions are made.

Dashboard Impact - The Power BI dashboard enabled interactive visualization of business performance, allowing stakeholders to compare spending patterns, regional performance, and profitability trends efficiently.

Business Recommendations

01 **Increase R&D Investment** - R&D showed the strongest positive impact on profit. Increasing R&D budget is the single most effective strategy to improve startup profitability.

02 **Maintain & Track Marketing Investment** - Maintain strategic marketing budgets; marketing expenditure showed a positive association with profit growth across startups.

03 **Optimize Administration Expenses** - Administration showed the weakest profit impact. Review and rationalize admin costs — eliminate inefficiencies and redirect resources to higher-impact activities.

04 **Leverage Predictive Analytics** - The regression model can estimate profit for new investment scenarios before committing budgets, enabling evidence-based financial decision-making.

05 **Benchmark NY Startups Practices** - New York recorded the highest cumulative profit. Studying and adopting successful practices from NY-based startups may benefit startups in other states.

06 **Avoid Indiscriminate Spending Increases** - More spending only helps when it produces measurable profit improvement. Focus resources on areas with proven returns, not across-the-board budget increases.

07 **Run ROI Checks for Major Budget Changes** - Before every significant budget decision, perform ROI analysis to prioritize investments that yield the highest return, maximizing resource efficiency.

CONCLUSION

Insights that Drive Profit

Profit Drivers Identified | **Model R^2=0.944** | **Actionable BI Insights**

This project successfully identified R&D, Marketing, Administration, and State as key profit drivers using multiple linear regression (R^2=0.944). Profit predictions were generated for two new scenarios, and all findings were visualized through an interactive Power BI dashboard — providing a strong, data-driven foundation for strategic investment and budgeting decisions.